COLLEGE OF ENGINEERING GUINDY CAMPUS

ANNA UNIVERSITY, CHENNAI

DEPARTMENT OF INFORMATION SCIENCE AND TECHNOLOGY (BATCH 1)

# DATA STRUCTURE PROJECT

## HOSTEL COMPLAINT PRIORITY MANAGEMENT SYSTEM USING DATA STRUCTURES IN C

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**IT23302 – DATA STRUCTURES**
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# Hostel Complaint Priority Management System Using Data Structures in C

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## 1. Problem Statement

In college hostels, students frequently face problems such as water supply issues, electrical faults, fan repairs, and plumbing problems. Managing these complaints manually can lead to delays, improper tracking, and difficulty in identifying urgent issues. Therefore, a computerized system is needed to register, organize, and prioritize hostel complaints efficiently.

## 2. Objectives

- To provide an efficient system for registering hostel complaints.
- To prioritize complaints based on their urgency.
- To reduce delays in resolving important maintenance issues.
- To maintain complaint records and track their status.
- To apply data structures such as Linked Lists and Queues using the C programming language.

## 3. Data Structures Used

- **Linked List:** Used to store complaint details dynamically, including complaint ID, student details, complaint category, priority, and status.
- **Queue:** Used to manage complaints in the order they are received when their priority levels are equal.
- **Searching:** Used to find complaints using their unique complaint ID.
- **Sorting:** Used to display complaints according to their priority level or registration date.

## 4. Enhancement Features

- Automatic complaint ID generation.
- Priority assignment based on complaint severity.
- Complaint status updates, such as Pending, In Progress, and Resolved.
- Search and filter options for complaint records.
- Generation of complaint summaries and resolution reports.
- File handling to preserve complaint records between program executions.

## 5. Working Flow

1. The user opens the system.
2. The user registers a new hostel complaint.
3. The system generates a unique complaint ID.
4. The complaint is assigned a priority based on its severity.
5. The complaint is stored using a Linked List.
6. Complaints are processed according to priority, with arrival order maintained for complaints of equal priority.
7. The administrator updates the complaint status.
8. The system displays pending and resolved complaints and generates a summary report.

## 6. Modules

- **Complaint Registration Module:** Collects and stores complaint details.
- **Priority Management Module:** Assigns priority levels based on complaint severity.
- **Complaint Processing Module:** Displays complaints in the appropriate processing order.
- **Search and Tracking Module:** Searches for complaints and checks their current status.
- **Admin Management Module:** Allows authorized staff to update complaint status and manage records.
- **Report Generation Module:** Displays summaries of pending, in-progress, and resolved complaints.

## 7. System Design

The system follows a modular design consisting of the following components:

- **Input Layer:** Accepts complaint details from the user.
- **Processing Layer:** Validates the input, assigns priorities, and manages complaints using a Linked List and Queue.
- **Search and Sorting Layer:** Retrieves complaint records and arranges them according to priority or date.
- **Output Layer:** Displays complaint details, processing order, status updates, and summary reports.
- **Storage Layer:** Uses file handling to save complaint records for future access.

The system is implemented in C using functions, structures, pointers, and data structures. It provides a simple and efficient approach to managing hostel maintenance complaints.